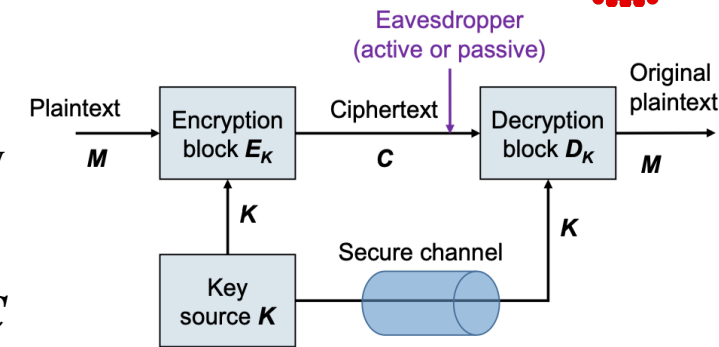


# Conventional Cryptography

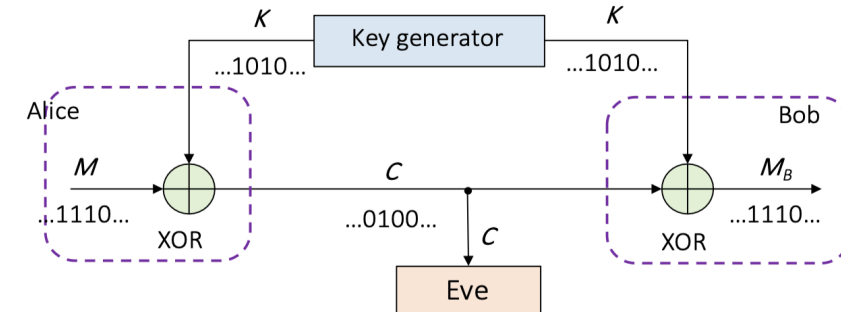


- Key Based Crypto systems:
  1. Symmetric Algorithms: Same key used for encryption & decryption
  2. Asymmetric Algorithms: Public encryption key & private decryption key
- Simplest private key cryptosystem: One time pad
- Unconditional (Information theoretic) Security: Message  $M$  and cryptogram  $C$  are statistically independent i.e.,  

$$I(M, C) = H(M) - H(M|C) = 0 \Leftrightarrow H(M|C) = H(M)$$
- Condition for perfect secrecy:  $H(M) \leq H(K)$
- Computational Security:
  1. Security is guaranteed when Eve runs cryptanalytic attack for limited amount of time. Security can be broken if sufficient computational resource and time are available.
  2. Eve can break the security with small probability of error.



Basic key-based cryptographic scheme

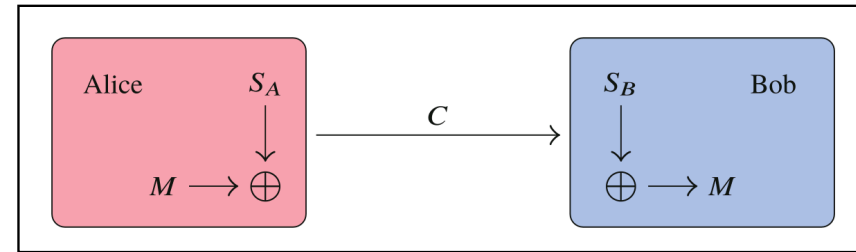


One-time pad encryption scheme



# One Time Pad

- Two parties, Alice and Bob, want to communicate securely.
- An adversary Eve can listen to everything but cannot alter messages.
- Alice and Bob share a secret random key  $S_A = S_B \in \{0,1\}^n$ .
- Message  $M = (M_1, \dots, M_n) \in \{0,1\}^n$ .



One-time pad encryption.

- **Goal:** Ensure Eve learns nothing about  $M$ .
- **Encryption:** Alice encrypts the message  $M$  using her key  $S_A$  :  $C = M \oplus S_A$ .
- **Transmission:** The ciphertext  $C$  is sent to Bob over a public channel.
- **Decryption:** Bob decrypts the message using his key  $S_B$  :  $M' = C \oplus S_B$ .

$$\begin{aligned} M'_i &= C_i \oplus (S_B)_i \\ &= (M_i \oplus (S_A)_i) \oplus (S_B)_i = M_i \oplus ((S_A)_i \oplus (S_B)_i) \\ &= M_i \oplus 0 \quad (\text{since } S_A = S_B) \\ &= M_i \end{aligned}$$

• **Correctness of Decryption:**



# One Time Pad

- **Perfect secrecy** proved by Claude Shannon (1949)
- **Requirements** for this protocol to be information-theoretically secure:
  - Key must be **truly random** (uncorrelated bits)
  - Key length  $\geq$  message length
  - Key must **never be reused**

$$C_1 = M \oplus S$$

- If same key  $S$  encrypts two messages  $M, N$ :

$$C_2 = N \oplus S$$

$$C_1 \oplus C_2 = M \oplus N$$

- Eve learns relationship between messages!
- Key must be **completely secret**
- **Limitations of One-Time Pad:**
  - Requires a truly random key  $S$  as long as the message.
  - Key must be **used only once** (hence the name).
  - Secure key distribution and storage is challenging.
  - Not practical for many modern applications.



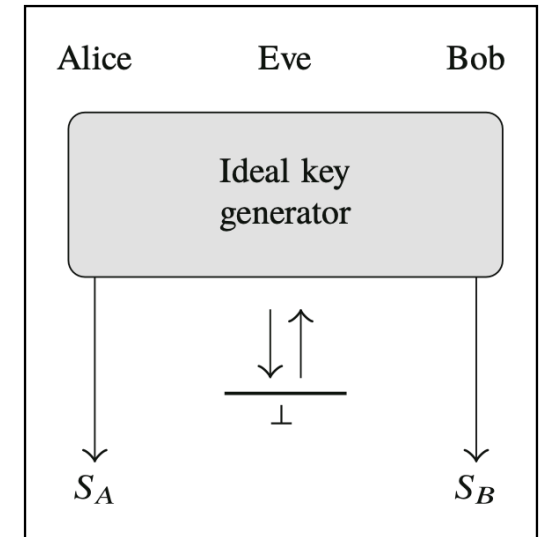
# Key Generation

- Ideal key generator setting.
- An ideal device outputs two identical, random keys  $S_A = S_B$  for Alice and Bob.
- Eve cannot interact with it (i.e., nothing can be input or output to Eve's part of the system, which is denoted by the symbol  $\perp$ ).
- Properties of Practical Key Generation:

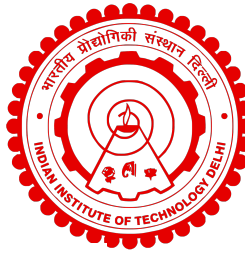
- **Correctness:** The probability that Alice and Bob's keys differ but the protocol **does not** abort is small:

$$\Pr[S_A \neq S_B] \leq \epsilon .$$

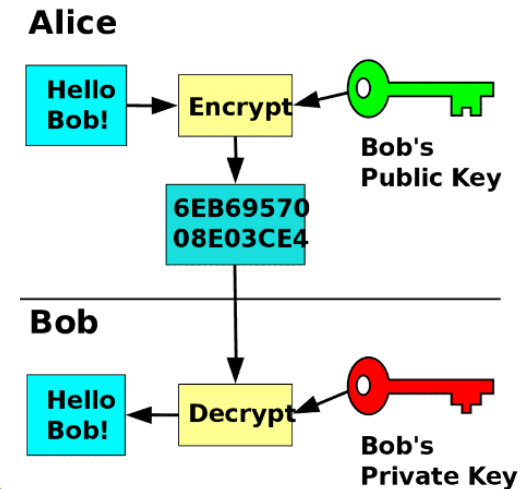
- **Close to perfect key:**
  - Eve does not have any knowledge of the key.
  - The individual key bits are uncorrelated.
- **Detectable attacks:** Protocol aborts if Eve interferes



# Current Security Issues



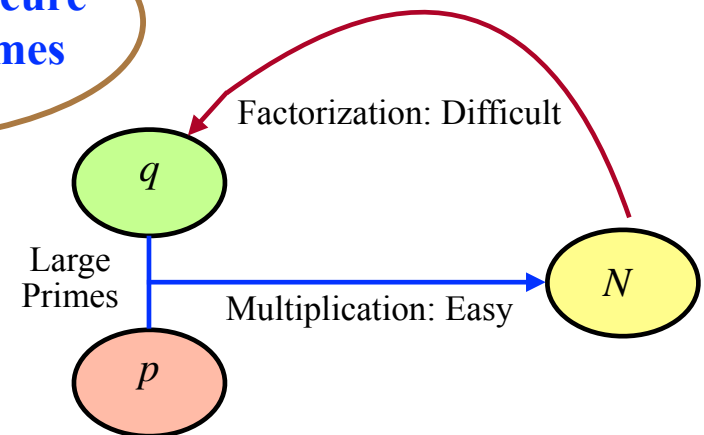
- **Aim:** Enhanced security and privacy.
- Public key encryption (RSA) is computationally secure
- Based on computationally hard problems of **prime factorization**
- Threat from **quantum computers**: Shor's factoring algorithm (polynomial complexity)



## Algorithms for Quantum Computation: Discrete Logarithms and Factoring

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600 Mountain Ave.  
Murray Hill, NJ 07974, USA

Need Quantum Secure  
Encryption Schemes



P. W. Shor, "Algorithms for quantum computation: Discrete logarithms and factoring," in *Proceedings 35th Annual Symposium on Foundations of Computer Science*, 1994.

P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM Rev*, vol. 41, (2), pp. 303-332, 1999.

# Quantum Secure Cryptography



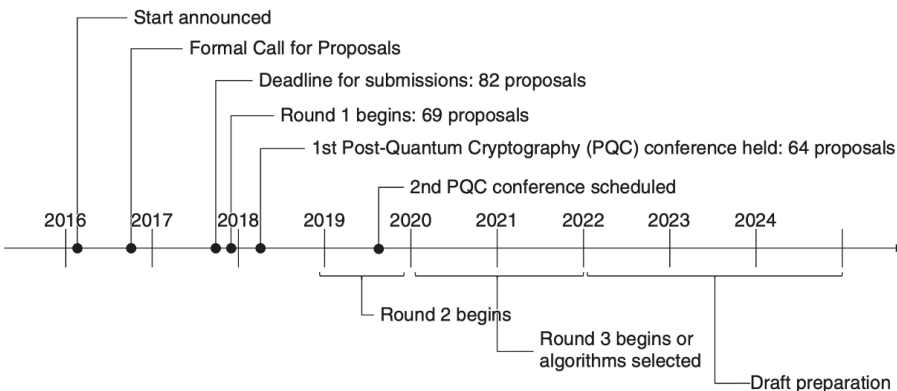
- Two Possible Solutions

## Post Quantum Cryptography (PQC)

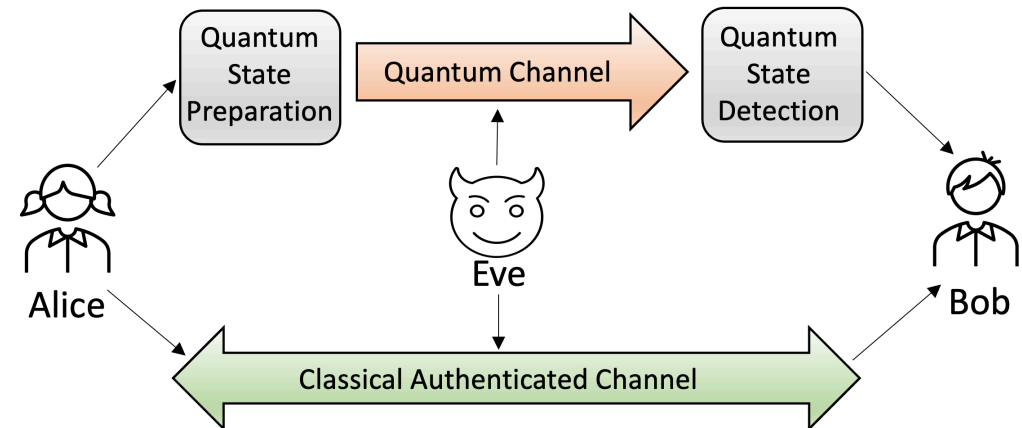
1. Harder computational problem.
2. A software solution
3. No security Proof
4. Eg. Lattice, Hash, Multivariate etc.

## Quantum Key Distribution (QKD)

1. Security guaranteed by quantum physics
2. Hardware solution
3. Unconditional security
4. Eg., DV-QKD , CV-QKD



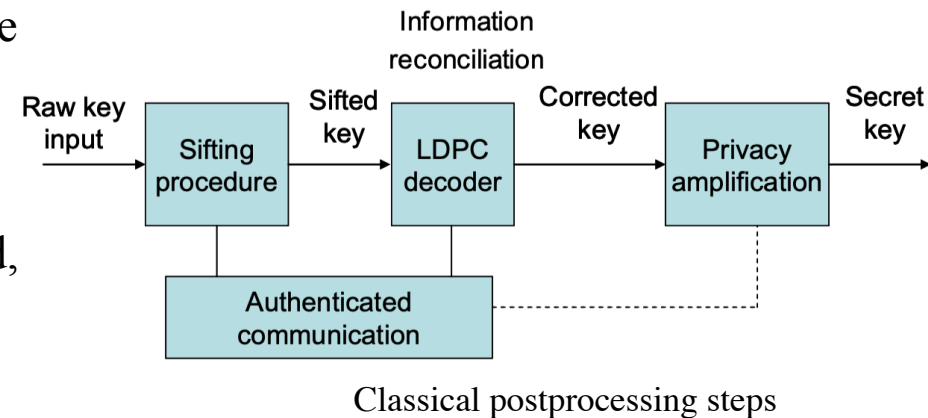
NIST Project Timeline





# Quantum Key Distribution

- QKD provides unconditional security guaranteed by the laws of quantum physics
  1. **No Cloning Theorem:** No quantum copier exists that can clone an arbitrary quantum state.
  2. **Indistinguishability of arbitrary quantum states:** It is impossible to unambiguously distinguish non-orthogonal quantum states. We studied POVM that minimises the error probability.
- Two categories of QKD protocols
  1. **Discrete Variable (DV-QKD):** Encodes the key information on single photon states (polarisation /phase)
  2. **Continuous Variable (CV-QKD):** Encodes the key information on quadratures of coherent/squeezed states. Security is based on Heisenberg's uncertainty principle.
- Since the raw key is imperfect, QKD also requires a classical authenticated channel for classical post-processing.
- **Information reconciliation:** Classical error correction to reconcile errors between  $X$  and  $Y$  to obtain a shared bit string  $K$  while divulging the information as little as possible to the Eve.
- **Privacy Amplification:** Distill from  $K$  a smaller set of bits  $S$  whose correlation with Eve's string  $Z$  is below a desired threshold, using universal hash function.



# QKD: Core Concept



- **Goal:** Let two parties (Alice and Bob) share a secure secret key over a public channel.
- **Security:** Guaranteed by quantum mechanics—not just technical constraints.
- **Implication:** Security holds unless quantum mechanics is disproven.
- **QKD Protocol Phases:**
  - **Quantum Transmission Phase:** Exchange or measure quantum states (qubits).
  - **Classical Post-Processing Phase:** Convert noisy raw data into a secure key using classical algorithms.